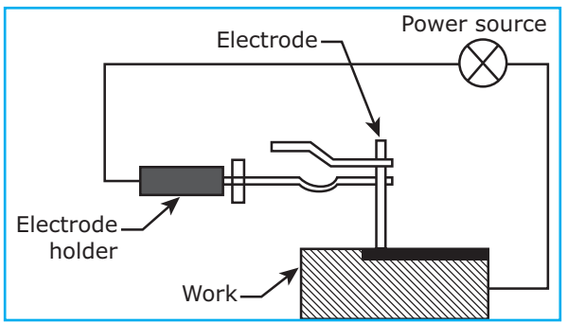


Arc welding equipments

1. D.C welding generator (or) AC transformer
2. Cables
3. Electrode cable
4. Work cable
5. Electrode holder
6. Electrode
7. Gloves
8. Protective shield (or) Goggles
9. Apron for the operator
10. Wire brush
11. Chipping hammer

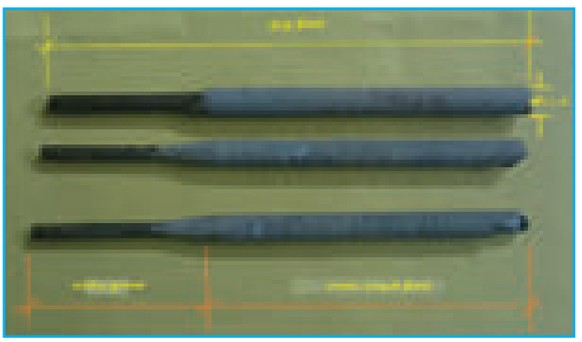
7.6 Differences between DC generator and AC transformer

DC generator	AC transformer
Low efficiency	High efficiency
High power cost	Low power cost
High machine cost	Low machine cost
Runs on low voltage	Runs on high voltage
Safe equipment	The chances of accidents is high
Uncoated electrodes may be used	Only flux coated electrodes are used
Joints are made with ferrous and non-ferrous metal parts.	Non-ferrous metal parts cannot be joined
Work is connected to the positive terminal (+) and the electrode is connected to the negative terminal (-)	No restriction in the direction of power supply connection
Noisy functioning	No noise during operation
Easy maintenance of equipments	Requires proper maintenance
The cost of the process is low	High process cost
Thin metal parts can easily be welded	Difficult to weld thin parts

7.7 Specification of electrodes

Electrodes - figure

Generally, the size of the electrodes is specified by the length and its diameter. They are available to a maximum of 12 mm diameter and 45mm length. The size of the electrode increases with the current used. In manual welding, the size of the electrode changes according to the thickness of the metal parts. Spring like electrodes are used in automatic welding.



7.8 Types of electrode

The electrodes used in arc welding are two types

1. Consumable electrode
2. Non – consumable electrode

XII BME_English version CHAPTER 07.indd 130 1/8/2020 1:40:24 PM